

Solutions second chapter the Standard model and Beyond

Abstract

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Useful formulas

A list of formulas used throughout the exercises.

$$s = E_{\text{CM}}^2 = 4(p^2 + m^2)$$

$$t = -2p^2(1 - \cos(\theta))$$

$$u = -2p^2(1 + \cos(\theta))$$

$$\phi_0 = \int [a(\vec{p})e^{-p \cdot x} + a^\dagger(\vec{p})e^{+ip \cdot x}] \frac{d^3\vec{p}}{(2\pi)^3 2E_p}$$

1 Exercise 1

The hamiltonian for a free complex scalar field is

$$H = \int \left[\left(\frac{\partial \phi^\dagger}{\partial t} \right) \left(\frac{\partial \phi}{\partial t} \right) + (\vec{\nabla} \phi^\dagger) \cdot (\vec{\nabla} \phi) + m^2 \phi^\dagger \phi \right] d^3 \vec{x}.$$

Show that

$$H = \int E_p \left[a^\dagger(\vec{p}) a(\vec{p}) + b^\dagger(\vec{p}) b(\vec{p}) \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E_p}$$

Assume that H is normal ordered.

To solve this exercise, we just need to compute everything using the field expression in function of the creators and annihilations operators. An example of such field is the ϕ_{π^+} which we take to annihilate a π^+ particle or, equivalently, to create a π^- . This means that $\phi_{\pi^+} = \phi_{\pi^-}^\dagger$.¹ With all this, the explicit definition of the fields are

$$\begin{aligned} \phi &= \int \left[a(\vec{p}) e^{-ip \cdot x} + b^\dagger(\vec{p}) e^{ip \cdot x} \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \\ \phi^\dagger &= \int \left[b(\vec{p}) e^{-ip \cdot x} + a^\dagger(\vec{p}) e^{ip \cdot x} \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \end{aligned}$$

Since we can go from ϕ to $\phi^\dagger$ by the substitution $a \mapsto b$, we compute the relevant quantities only for ϕ . Just so that we may speed up our calculations,

$$\begin{aligned} \frac{d}{dt} e^{-ip \cdot x} &= -iE_p e^{-ip \cdot x} \\ \frac{d}{dt} e^{ip \cdot x} &= iE_p e^{ip \cdot x} \\ \vec{\nabla} e^{-ip \cdot x} &= i\vec{p} e^{-ip \cdot x} \\ \vec{\nabla} e^{ip \cdot x} &= -i\vec{p} e^{ip \cdot x} \end{aligned}$$

Onwards!

¹It is in this sense that we might say that creating a particle and destroying an antiparticle is the same operation, although the two have opposite 4-momenta.

$$\begin{aligned}
\frac{d\phi}{dt} &= \int \left[a \frac{d}{dt} e^{-ip \cdot x} + b^\dagger \frac{d}{dt} e^{+ip \cdot x} \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E} = \int [-iE a e^{-ip \cdot x} + iE b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{d(2\pi)^3 2E_p} \\
&= \int iE [-a e^{-ip \cdot x} + b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{d(2\pi)^3 2E} \\
\vec{\nabla} \phi &= \int [i p a e^{-ip \cdot x} - i \vec{p} b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{(2\pi)^3 2E} = \int i \vec{p} [a e^{-ip \cdot x} - b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{(2\pi)^3 2E}
\end{aligned}$$

Therefore, by multiplying the correspond quantites of $\frac{d\phi^\dagger}{dt}$ and $\nabla \phi^\dagger$ we obtain

$$\begin{aligned}
H &= \int \left[\left(\frac{\partial \phi^\dagger}{\partial t} \right) \left(\frac{\partial \phi}{\partial t} \right) + (\vec{\nabla} \phi^\dagger) \cdot (\vec{\nabla} \phi) + m^2 \phi^\dagger \phi \right] d^3 \vec{x} \\
&= \int - (EE' + |\vec{p}| |\vec{p}'|) [a b e^{-2(\dots)} - a^\dagger a - b^\dagger b + a^\dagger b^\dagger e^{+2(\dots)}] \frac{d^3 \vec{p}}{(2\pi)^3 2E} d^3 \vec{x} \\
&\quad + \int m^2 [a b e^{-2(\dots)} + a^\dagger a + b^\dagger b + b^\dagger a^\dagger e^{+2(\dots)}] \frac{d^3 \vec{p}}{(2\pi)^3 2E} \frac{d^3 \vec{p}'}{(2\pi)^3 2E} d^3 \vec{x}
\end{aligned}$$

where $(\dots) = i(p + p') \cdot x$. These are a lot of terms, so something ought to cancel. Remember that we still have an integration to carry out, and this is explicitly

$$\int e^{\pm i(p+p') \cdot x} d^3 \vec{x} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}')$$

This means that we need to put $\vec{p} = \vec{p}'$. The last ingredient is using the on-mass shell prescription, which tells that we need to consider only the 4-momenta with $p^2 = m^2$.

The only terms which survive are those without an $e^{(\dots)}$, that is

$$H = \int 2E^2 [a^\dagger a + b^\dagger b] \frac{d^3 \vec{p}}{(2\pi)^3 2E^2} = \int E [a^\dagger a + b^\dagger b] \frac{d^3 \vec{p}}{(2\pi)^3 2E}$$

Now, let's see if this result make physically sense. First of all, we can count the mass-powers in the result and see that the whole has $[H]_m = 1$, which is good since it has to be an energy. Tracing back the definitions, remebering that $[\phi]_m = 1$, we have that the operator a, b have mass dimension of -1 , thereby showing that $[H]_m = 1$.

2 Exercise 2

The CM differential and total cross sections for the elastic scattering of a Hermitian scalar of mass m with potential $V_I(\phi) = k \frac{\phi^3}{3!} + \lambda \frac{\phi^4}{4!}$ are given by the formula

$$\frac{d\sigma}{d\cos(\theta)} = \frac{1}{32\pi s} \left| \lambda + k^2 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right] \right|^2$$
$$\sigma = \frac{1}{2} \int_{-1}^1 \frac{d\sigma}{d\cos(\theta)} d\cos(\theta)$$

Choose $m = 0.5\text{GeV}$, $\lambda = 0.3$ and $k = 1.2\text{GeV}$. Plot the CM differential cross section $\frac{d\sigma}{d\cos(\theta)}$ in units of $\text{fm}^2 = 10^{-26}\text{cm}^2$ as a function of $\cos(\theta)$ for $s = 5m^2, 6m^2$, and $7m^2$, and plot the total cross section σ as a function of $\sqrt{s}$ for $2m < \sqrt{s} < 3m$.

First of all, we choose s and $\cos(\theta)$ as independent variables, so that

$$s = 4(p^2 + m^2)$$
$$t = -2 \left(\frac{2}{4} - m^2 \right) (1 - \cos(\theta))$$
$$u = -2 \left(\frac{2}{4} - m^2 \right) (1 + \cos(\theta))$$

3 Exercise 3

Derive the equation

$$\frac{d\sigma}{d\cos(\theta_3)} \stackrel{!}{=} \frac{|M_{fi}|^2}{32\pi p_1 m_2} \frac{p_3}{E_1 + m_2 - \frac{p_1 E_3}{p_3} \cos(\theta_3)}$$

, where the θ_3 refers to the laboratory scattering angle of particle 3, for the special case $m_1 = m_3 = 0$ and $m_2 = m_4$ by Lorentz transforming the equation

$$\frac{d\sigma}{d\cos(\theta)} = \frac{1}{32\pi s} \frac{p_f}{p_i} |M_{fi}|^2$$

We may use the fact that σ, s and t are Lorentz's invariants.

The exercise asks to go from the center of momentum frame (or center of mass) straight into the laboratory frame. Therefore, we need to check how $\cos(\theta), p_i$ and p_f change as we go from one coordinate system to another.

So, let's write what we know about the CM frame. Knowing that in this frame the total three momentum cancels (hence the name), we have

$$\begin{aligned} p_1 &= (E_1 \quad \vec{p}_i) \\ p_2 &= (E_2 \quad -\vec{p}_i) \end{aligned}$$

Now, to go from this frame to the lab one, that is to

$$\begin{aligned} p_1 &= (E_1^{\text{lab}} \quad \vec{p}_1) \\ p_2 &= (E_2^{\text{Lab}} \quad \vec{p}_2) \end{aligned}$$

we use the relativistic Lorentz addition imposing $\vec{\beta} = \frac{\vec{p}_i}{E_2}$. With this requirement, we get that $\vec{p}_2 = \vec{0}$, since²

$$\begin{aligned} \vec{p}_2 &= \gamma_L(-\vec{p}_i + \vec{\beta}_L E_2) \\ &= \gamma_L(-\vec{p}_i + \vec{p}_i) = 0 \end{aligned}$$

Without further ado, let's compute what are the 4-momentas p_1 and p_3 in the new frame.

²In this example, I'm tacitly assuming that all the computations are in collinear frames, that is $\vec{p}_i = (|\vec{p}_i| \quad 0)$

$$\begin{aligned}
E_1^{\text{Lab}} &= \frac{E_2}{m_2} \left(E_1 + \frac{1}{E_2} |\vec{p}_i|^2 \right) \\
&= \frac{E_2 E_1}{m_2} + \frac{1}{m_2} (E_2^2 - m_2^2) = \frac{E_2 \sqrt{s}}{m_2} - m_2 \\
p_1^{\vec{\text{Lab}}} &= \frac{E_2}{m_2} \left(1 + \frac{E_1}{E_2} \right) \vec{p}_i = \frac{E_1 + E_2}{m_2} \vec{p}_i = \frac{\sqrt{s}}{m_2} \vec{p}_i \\
E_3^{\text{Lab}} &= \frac{E_2 E_3}{m_2} + \frac{1}{m_2} |\vec{p}_i| |\vec{p}_f| \cos(\theta) \\
\vec{p}_3^{\text{Lab}} &= \left(\frac{E_2}{m_2} |\vec{p}_f| \cos(\theta) + \frac{E_3}{m_2} |\vec{p}_i| \right) \vec{p}_i + |\vec{p}_f| \sin(\theta) \vec{p}_\perp
\end{aligned}$$

Since we have computed all the quantities in the new frame, I will call them $E_1, E_3, \vec{p}_1$ and $\vec{p}_3$ without the superscript "Lab". The last expression allows us to get how the new angle of scattering θ_3 is related to the old scattering angle θ . To see this, we name the components of $\vec{p}_3$ as $\vec{p}_3 = (|\vec{p}_3| \cos(\theta_3) \quad |\vec{p}_3| \sin(\theta_3))$ so that

$$\begin{cases} |\vec{p}_3| \cos(\theta_3) = \frac{E_2}{m_2} |\vec{p}_f| \cos(\theta) + \frac{E_3}{m_2} |\vec{p}_i| \\ |\vec{p}_3| \sin(\theta_3) = |\vec{p}_f| \sin(\theta) \end{cases}$$

Since the two expressions above are simultaneously valid, we can take the first equation and divide it by the second, getting

$$\tan(\theta_3(\theta)) = \frac{|\vec{p}_f| \sin(\theta)}{\frac{E_2}{m_2} |\vec{p}_f| \cos(\theta) + \frac{E_3}{m_2} |\vec{p}_i|}$$

Our objective is to compute the differential cross section in the new frame. Therefore, we need to change variables from $\theta \mapsto \theta_3$, that is

$$\frac{d\sigma}{d\cos(\theta_3)} \frac{d\cos(\theta_3)}{d\cos(\theta)}$$

Luckily, our work from before may be used precisely for this. First of all, we have $\tan(\theta_3)$ and not $\cos(\theta_3)$ as a function of the old variables. But, we can employ a trigonometric trick to rewrite³

$$\cos(\theta_3) = \frac{\cos(\theta_3)}{\sqrt{\cos^2(\theta_3) + \sin^2(\theta_3)}} = \frac{1}{\sqrt{1 + \tan^2(\theta_3)}}.$$

³Here we are interested only in the magnitude of the derivative, so we may as well suppose that $\cos(\theta_3)$ is positive.

Furthermore,

$$\begin{aligned}
\frac{d\cos(\theta_3)}{d\cos(\theta)} &= \frac{d\theta}{d\cos(\theta)} \frac{d\cos(\theta_3)}{d\theta} = -\frac{1}{\sin(\theta)} \frac{d\cos(\theta_3)}{d\theta} \\
&= -\frac{1}{\sin(\theta)} \frac{d}{d\theta} \left(\frac{1}{\sqrt{1 + \tan^2(\theta_3(\theta))}} \right) \\
&= \frac{1}{\sin(\theta)} \frac{\tan(\theta_3)}{\sqrt{1 + \tan^2(\theta_3)}} \\
&= \frac{\sin(\theta_3)}{\sin(\theta)} = \frac{|\vec{p}_f|}{|\vec{p}_3|}.
\end{aligned}$$

With all this, we just turn to the differential cross section⁴,

$$\frac{d\sigma}{d\cos(\theta_3)} = \frac{d\sigma}{d\cos(\theta)} \frac{d\cos(\theta)}{d\cos(\theta_3)} = \frac{1}{32\pi |\vec{p}_1| m_2} \frac{|M_{fi}|^2}{\sqrt{s}}$$

Now, we just need to specialize to the case prescribed, that is where $m_1 = 0 = m_3$ and $m_2 = m_4$. From our general formula, we substitute $E_3 = |\vec{p}_3|$, $E_1 = |\vec{p}_1|$. Furthermore, we know that in this special case

$$\frac{p_1}{p_3} = \frac{1}{1 + \frac{p_1}{m_2}(1 - \cos(\theta_3))}$$

$$\begin{aligned}
\frac{d\sigma}{d\cos(\theta_3)} &\stackrel{\text{Special case}}{=} \frac{|M_{fi}|^2}{32\pi p_1 m_2} \frac{p_3}{|\vec{p}_1| + m_2 - |\vec{p}_1| \cos(\theta_3)} \\
&= \frac{|M_{fi}|^2}{32\pi |\vec{p}_1| m_2} \frac{p_3}{p_1 m_2 \left(1 + \frac{|\vec{p}_1|}{m_2} \cos(\theta_3)\right)} \\
&= \frac{|M_{fi}|^2}{32\pi m_2^2} \left(\frac{|\vec{p}_3|}{|\vec{p}_1|} \right)^2
\end{aligned}$$

⁴I currently lack the proof that $\sqrt{s} = E_1 + m_2 - \frac{|\vec{p}_1| E_3}{|\vec{p}_3|} \cos(\theta_3)$.

4 Exercise 4

Consider the process $\pi^+(\vec{p}_1)\pi^-(\vec{p}_2) \rightarrow \pi^+(\vec{p}_3)\pi^-(\vec{p}_4)$ with $p_1 \neq p_3$ and $p_2 \neq p_4$, in the theory of a complex scalar field with Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m^2 \phi^\dagger \phi - V_I(\phi, \phi^\dagger) = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m^2 \phi^\dagger \phi - \frac{\lambda}{4} (\phi^\dagger \phi)^2.$$

The tree-level amplitude M_{fi} is

$$(2\pi)^4 \delta^4(p_3 + p_4 - p_1 - p_2) M_{fi} = \int \langle \vec{p}_3 \vec{p}_4 | -iV_I(\phi_0(x), \phi_0^\dagger(x)) | \vec{p}_1 \vec{p}_2 \rangle d^4x$$

Calculate the amplitude using the free-field expression for ϕ_0 , and show that $M_{fi} = -i\lambda$

Before diving into this exercise, let's see what are the states we are working with. The process we describe is $\pi^+(\vec{p}_1)\pi^-(\vec{p}_2) \rightarrow \pi^+(\vec{p}_3)\pi^-(\vec{p}_4)$. As we have done back in exercise 1, let's decide that ϕ_π destroys π^+ particles while $\phi^\dagger$ destroys them. According to this convention, we have

$$\begin{aligned} a^\dagger(\vec{p}) |0\rangle &= |\pi^+(\vec{p})\rangle \\ b^\dagger(\vec{p}) |0\rangle &= |\pi^-(\vec{p})\rangle \end{aligned}$$

Then, we see that

$$|\vec{p}_1 \vec{p}_2\rangle = |\pi^+(\vec{p}_1)\pi^-(\vec{p}_2)\rangle = a^\dagger(\vec{p}_1)b^\dagger(\vec{p}_2) |0\rangle$$

Now, let's consider the product of these 4 fields. When we evaluate them, we get a combination of $a, b, a^\dagger, b^\dagger$ which either creates or destroy particles or antiparticles. Within these 16 different terms, what should we keep? The $\pi^\pm$ is a scalar field, so the final state can have multiple bosons, but here we are interested in the scalar product with another 2-particle state, that is $\langle \pi^+(\vec{p}_3)\pi^-(\vec{p}_4) |$. Therefore, the combination of $a, b, a^\dagger, b^\dagger$ that we keep are only those which leave the final number of particles intact, that is 1 particle and 1 antiparticle. To compactify the notation, call

$$\begin{aligned} A_{pq}^{lm} &\equiv \frac{1}{(2\pi)^{3(l+m+p+q)}} \frac{1}{(2E_{p_i})^l (2E_{p_j})^m (2E_{p_k})^p (2E_{p_n})^q} \times \\ &\quad (a^\dagger(\vec{p}_a))^l e^{+i\left(\sum_{k=1}^l p_k\right) \cdot x} a^m(\vec{p}_b) e^{-i\left(\sum_{k=1}^m p_k\right) \cdot x} b^p(\vec{p}_c) e^{-i\left(\sum_{k=1}^p p_k\right) \cdot x} (b^\dagger(\vec{p}_d))^q e^{+i\left(\sum_{k=1}^q p_k\right) \cdot x} \\ &= \frac{1}{(2\pi)^{3(l+m+p+q)}} \frac{1}{(2E_{p_i})^l (2E_{p_j})^m (2E_{p_k})^p (2E_{p_n})^q} \times \\ &\quad (a^\dagger(\vec{p}_a))^l a^m(\vec{p}_b) b^p(\vec{p}_c) (b^\dagger(\vec{p}_d))^q e^{+i\left(\sum_{i,j,k,n=1}^{l,m,p,q} (p_i + p_n - p_j - p_k)\right) \cdot x} \end{aligned}$$

The only A 's which survive are A_{11}^{11} , A_{00}^{22} or A_{22}^{00} . We have only 1 way to get A_{00}^{22} and A_{22}^{00} , whereas we have 4 ways to write down A_{11}^{11} .

Since we have to integrate over $\int d^4x$, we see that

$$\int A_{pq}^{lm} d^4x = \text{Polynomial}(a^\dagger, a, b, b^\dagger) \delta^4\left(\sum_{i,j,k,n=1}^{l,m,p,q} (p_i + p_n - p_j - p_k)\right)$$

Let's compute explicitly the term A_{00}^{22} . To do so, we use the commutation rule for the complex scalar field. To reduce the notation, define $a_{\vec{p}_{\text{any}}} \equiv a_{\text{any}}$.

$$\begin{aligned} \langle \vec{p}_3 \vec{p}_4 | a_a a_b a_c a_d | \vec{p}_1 \vec{p}_2 \rangle &= \langle 0 | a_3 b_4 a_a a_b a_c a_d a_1^\dagger b_2^\dagger | 0 \rangle = \langle 0 | a_3 a_a a_b a_c a_d a_1^\dagger b_4 b_2^\dagger | 0 \rangle \\ &= (2\pi)^3 2E_{p_4} \delta^{(3)}(\vec{p}_4 - \vec{p}_2) \langle 0 | a_3 a_a a_b a_c a_d a_1^\dagger | 0 \rangle \end{aligned}$$

That last element is 0.! We can either compute it directly

$$\langle 0 | a_3 a_a a_b a_c a_d a_1^\dagger | 0 \rangle = (2\pi)^5 2^5 E_3 E_a E_b E_c E_d \delta^{(3)}(\dots) \langle 0 | a_1^\dagger a_3 a_a a_b a_c a_d | 0 \rangle = 0$$

or we recognize that that element describes the transition

$$\langle 0 | a_3 a_a a_b a_c a_d a_1^\dagger | 0 \rangle = \langle \vec{p}_3 \vec{p}_a \vec{p}_b \vec{p}_c \vec{p}_d | \vec{p}_1 \rangle = \langle \pi^+ \pi^+ \pi^+ \pi^+ \pi^+ | \pi^+ \rangle$$

which is 0 since we can't obtain 5 π^+ from a single one!⁵

Now we compute (one of) the other A_{11}^{11} terms, getting

$$\begin{aligned} \langle 0 | a_3 b_4 a_a^\dagger a_b b_c^\dagger a_1^\dagger b_2^\dagger | 0 \rangle &= \left\langle 0 \left| a_3 a_a^\dagger a_b a_1^\dagger b_4 b_c^\dagger b_2^\dagger \right| 0 \right\rangle \\ &= (2\pi)^6 2E_{p_3} 2E_{p_1} \delta^{(3)}(\vec{p}_3 - \vec{p}_a) \delta^{(3)}(\vec{p}_1 - \vec{p}_b) \left\langle 0 \left| b_4 b_c^\dagger b_2^\dagger \right| 0 \right\rangle \end{aligned}$$

Focusing on the last term, we get all the contributions from inverting the momenta of the π^- particles, that is

$$\begin{aligned} \left\langle 0 \left| b_4 b_c^\dagger b_2^\dagger \right| 0 \right\rangle &= (2\pi)^3 2E_{p_c} \delta^{(3)}(\vec{p}_c - \vec{p}_d) \left\langle 0 \left| b_4 b_2^\dagger \right| 0 \right\rangle + \left\langle 0 \left| b_4 b_d^\dagger b_c b_2^\dagger \right| 0 \right\rangle \\ &= (2\pi)^6 2E_{p_c} 2E_{p_4} \delta^{(3)}(\vec{p}_4 - \vec{p}_2) \delta^{(3)}(\vec{p}_c - \vec{p}_d) + \\ &\quad + (2\pi)^6 2E_{p_d} 2E_{p_c} \delta^{(3)}(\vec{p}_d - \vec{p}_4) \delta^{(3)}(\vec{p}_c - \vec{p}_2) \end{aligned}$$

Why did we get two contributions from the b terms and only one from the a terms? Notice that the operator order is such that we can't invert their momenta, that is we do have a way to distinguish them. Of course, if this argument doesn't suffice we can always compute them. Putting all together, we get that

$$\begin{aligned} \left\langle \vec{p}_3 \vec{p}_4 \left| a_a^\dagger a_b b_c b_d^\dagger \right| \vec{p}_1 \vec{p}_2 \right\rangle &= (2\pi)^{12} 2E_{p_3} 2E_{p_1} \delta^{(3)}(\vec{p}_3 - \vec{p}_a) \delta^{(3)}(\vec{p}_1 - \vec{p}_b) \times \\ &\quad \times (2E_{p_c} 2E_{p_4} \delta^{(3)}(\vec{p}_4 - \vec{p}_2) \delta^{(3)}(\vec{p}_c - \vec{p}_d) + 2E_{p_d} 2E_{p_c} \delta^{(3)}(\vec{p}_d - \vec{p}_4) \delta^{(3)}(\vec{p}_c - \vec{p}_2)) \end{aligned}$$

⁵Notice that we can't even if we launch the initial pion at momentum, in mass units, $p_{\pi, \text{in}} \approx 5m_\pi$, since the process is prohibited by number conservation!

This is one of 4 contributions, which we see have the peculiarity of conserving the 4 momenta of the inner particles. Do not forget about the $\delta^{(3)}(\vec{p}_a + \vec{p}_b - \vec{p}_c - \vec{p}_d)$ that we got by integrating over space! That is crucial, since all the deltas we obtained above can be, ultimately, written in terms of the physical momenta. Therefore, we claim that

$$\langle \vec{p}_3 \vec{p}_4 | A_{11}^{11} | \vec{p}_1 \vec{p}_2 \rangle = 4(2\pi)^4 (2\pi)^{12} 2E_{p_a} 2E_{p_b} 2E_{p_c} 2E_{p_d} \delta^3(\text{inner momenta}) \delta^{(4)}(p_1 + p_2 - p_3 - p_4)$$

With all this,

$$\begin{aligned} \int \langle \vec{p}_3 \vec{p}_4 | \phi_0^2 (\phi_0^\dagger)^2 | \vec{p}_1 \vec{p}_2 \rangle d^4x &= \int \langle \vec{p}_3 \vec{p}_4 | A_{11}^{11} | \vec{p}_1 \vec{p}_2 \rangle \frac{d^3\vec{p}_a}{d(2\pi)^3 2E_{p_a}} \frac{d^3\vec{p}_b}{d(2\pi)^3 2E_{p_b}} \frac{d^3\vec{p}_c}{d(2\pi)^3 2E_{p_c}} \frac{d^3\vec{p}_d}{d(2\pi)^3 2E_{p_d}} \\ &= 4(2\pi)^4 \delta^{(3)}(p_1 + p_2 - p_3 - p_4) \times \\ &\times \int \delta^{(3)}(\text{inner momenta}) d^3(\text{inner momenta}) = \\ &= 4(2\pi)^4 \delta^{(3)}(p_1 + p_2 - p_3 - p_4) \end{aligned}$$

So,

$$(2\pi)^4 M_{fi} \delta^{(p_1 + p_2 - p_3 - p_4)} = -i \frac{\lambda}{4} (2\pi)^4 \cdot 4(2\pi)^4 \delta^{(3)}(p_1 + p_2 - p_3 - p_4),$$

therefore

$$M_{fi} = -i\lambda.$$

5 Exercise 5

Consider the following Lagrangian density

$$\mathcal{L} = \sum_{i=1}^3 \frac{1}{2} [(\partial_\mu \phi_i)^2 - m_i^2 \phi_i^2] - k \phi_1 \phi_2 \phi_3$$

for three non-identical Hermitian fields. Calculate the lowest order differential cross section in the center of mass for $\phi_1(\vec{p}_1)\phi_2(\vec{p}_2) \rightarrow \phi_1(\vec{p}_3)\phi_2(\vec{p}_4)$ as a function of s and the CM scattering angle θ for the special case $m_1 = m_2 \neq m_3$.

The differential cross section for a $2 \rightarrow 2$ process, in the CM, is

$$\frac{d\sigma}{d\cos(\theta)} = \frac{1}{32\pi s} \frac{p_f}{p_i} |M_{fi}|^2$$

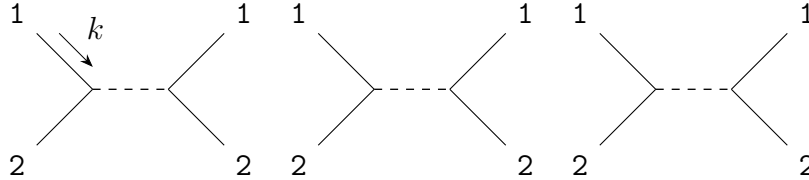
Let's deal with the kinematics first. Using the energy conservation, keeping the notation implemented in exercise 3, we get

$$\begin{aligned} E_{1,i}^2 + E_{2,i}^2 + 2E_{1,i}E_{2,i} &= E_{1,f}^2 + E_{2,f}^2 + 2E_{1,f}E_{2,f} \\ m_1^2 + m_2^2 + 2p_1^2 + 2\sqrt{p_i^2 + m_1^2}\sqrt{p_i^2 + m_2^2} &= m_1^2 + m_2^2 + 2p_f^2 + \sqrt{p_f^2 + m_1^2}\sqrt{p_f^2 + m_2^2} \\ p_i^2 + \sqrt{p_i^2 + m_1^2}\sqrt{p_i^2 + m_2^2} &= p_f^2 + \sqrt{p_f^2 + m_1^2}\sqrt{p_f^2 + m_2^2} \end{aligned}$$

where we write $|p_{\text{any}}^\rightarrow|^2 \equiv p_{\text{any}}^2$. Now we specialize the above for the case $m_1 \equiv m \equiv m_2$, obtaining

$$2p_i^2 + m^2 = 2p_f^2 + m^2 \implies p_i^2 = p_f^2$$

which is to say $|\vec{p}_i| = \pm |\vec{p}_f|$. How do we interpret this result? Being initial and final momenta equal, we know this to be an elastic scattering! Finally, we turn to the interactions. The lowest order relevant diagrams are the s, t and u channel, that is ⁶



Now we use the Feynmann rules and write out the resulting M_{fi} . Using the above diagrams, we see that

$$M_{fi} = (-ik)^2 \left[\frac{1}{s - m_3^2} + \frac{1}{t - m_3^2} + \frac{1}{u - m_3^2} \right]$$

⁶I'm tacitly assuming time to go from left to right in this page.

The Feynmann diagrams allows us to appreciate the fact that the interaction between particle 1 and 2 is mediated by particle 3.⁷

Now, it is time to piece everything together. We have seen how the particles do an elastic scattering. Therefore, in this case,

$$\begin{aligned}t &= -2p^2(1 - \cos(\theta)) \\ u &= -2p^2(1 + \cos(\theta))\end{aligned}$$

The amplitude is equal to

$$M_{fi} = (-ik)^2 \left[\frac{1}{s - m_3^2} + \frac{1}{-2p^2(1 - \cos(\theta)) - m_3^2} + \frac{1}{-2p^2(1 + \cos(\theta)) - m_3^2} \right]$$

So, we draw the conclusion that

$$\frac{d\sigma}{d\cos(\theta)} = \frac{k^2}{32\pi s} \left| \left[\frac{1}{s - m_3^2} + \frac{1}{-2p^2(1 - \cos(\theta)) - m_3^2} + \frac{1}{-2p^2(1 + \cos(\theta)) - m_3^2} \right] \right|^2$$

Notice that we need to keep the absolute value since t and u may be negative at high energies.

⁷Notice that the particle 3 in this process is a virtual particle since it is not an observable: it doesn't appear either in the initial nor in the final state.

6 Exercise 6

Suppose that the interaction potential for a complex scalar field were

$$V_I = \sigma_4(\phi^4 + \phi^{4,\dagger}),$$

(rather than $\lambda \frac{(\phi\phi^\dagger)^2}{4}$), which is not $U(1)$ invariant. Show that the charge is not conserved and calculate the lowest order amplitude for $\pi^+(\vec{p}_1)\pi^+(\vec{p}_2) \rightarrow \pi^-(\vec{p}_3)\pi^-(\vec{p}_4)$.

7 Exercise 7

Consider charged pion QED with a massive photon, i.e. add the term $\frac{1}{2}M_A^2 A_\mu A^\mu$ to the lagrangian density (insert qed lagrangian) (with $V_I = 0$). assume $M_A > 2m$.⁸

1. Calculate the decay rate for $\gamma \rightarrow \pi^+ \pi^-$ in the photon rest frame at tree level for an unpolarized massive photon.
 2. Calculate the π^+ angular distribution $\frac{d\Gamma}{d\cos(\theta)}$ for a polarized photon, where θ is the angle between the photon polarization direction in the rest frame and the π^+ direction.
 3. Show that one recovers the result in 1. when $\frac{d\Gamma}{d\cos(\theta)}$ is integrated over $\cos(\theta)$.
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⁸The photon mass term is not gauge invariant, but the model still makes sense at tree level

8 Exercise 8

Consider $\pi^+(\vec{p}_1)\pi^-(\vec{p}_2) \rightarrow \pi^+(\vec{p}_3)\pi^-(\vec{p}_4)$ in massive scalar electrodynamics, i.e. with

$$V_I = g\phi^\dagger,$$

where A , the analog of the electromagnetic field, is a Hermitian spin-0 field with mass $\mu \neq 0$. The analog of the charge is g , which now has dimension of mass.

1. Find expressions for the differential and total cross sections in the CM to lowest non-trivial order, in terms of s, m, μ, g and $\cos(\theta)$.
 2. Define the dimensionless variable $x = \frac{s}{m^2} \geq 4$, and specialize to the values $m = 1\text{GeV} = g$, $\mu = 0.5\text{GeV}$. Plot $\frac{d\sigma}{d\cos(\theta)}$ in units of $fm^2 = 10^{-26}\text{cm}^2$ for $x = 4, 4.2$ and 4.4 .
 3. Plot σ in units of fm^2 versus x for the same parameter values and the range $4 \leq x \leq 5$.
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9 Exercise 9

Prove directly from the defining relations that

$$\sigma^{\mu\nu}\gamma^5 = -\frac{i}{2}\epsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma}.$$

10 Exercise 10

Prove the Gordon decomposition formulas

$$\begin{aligned}2m(\bar{u}_2\gamma^\mu u_1) &= \bar{u}_2(p_1 + p_1)^\mu u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 - p_1)_\nu u_1 \\2m(\bar{u}_2\gamma^\mu\gamma^5 u_1) &= \bar{u}_2(p_2 - p_1)^\mu\gamma^5 u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 + p_1)_\nu\gamma^5 u_1 \\0 &= \bar{u}_2(p_2 - p_1)^\mu u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 + p_1)_\nu u_1 \\0 &= \bar{u}_2(p_1 + p_1)^\mu\gamma^5 u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 - p_1)_\nu\gamma^5 u_1\end{aligned}$$

where $u_{1,2}$ are two Dirac u spinors for a particle of mass m . Damn

11 Exercise 11

Prove the identity

$$\gamma^\mu \gamma^\nu \gamma^\rho = \gamma^\mu g^{\nu\rho} + \gamma^\rho g^{\mu\nu} - \gamma^\nu g^{\mu\rho} + i\epsilon^{\sigma\mu\nu\rho} \gamma_\sigma \gamma^5$$

12 Exercise 12

Show by explicit construction that the Pauli-Dirac and chiral representations are related by a unitary transformation, i.e. that there exists a unitary matrix U such that

$$\begin{aligned}U\gamma_{\text{PD}}^\mu U^\dagger &= \gamma_{\text{ch}}^\mu \\Uu_{\text{PD}}(\vec{p}, s) &= u_{\text{ch}}(\vec{p}, s) \\Uv_{\text{PD}}(\vec{p}, s) &= -v_{\text{ch}}(\vec{p}, s)\end{aligned}$$

The extra sign in the v -spinor transformation is due to a sign convention.

13 Exercise 13

The angular momentum operator for a Dirac field can be written as

$$J^i = \int \psi^\dagger(x) \frac{1}{4} \epsilon^{ijk} \sigma_{jk} \psi(x) \, d^3\vec{x} + \text{orbital}$$

where normal ordering is implied. Show that, in the free field limit, J^3 has the expected behaviour, that is

$$J^3 |\psi(\vec{p}, s_{1,2})\rangle = \pm \frac{1}{2} |\psi(\vec{p}, s_{1,2})\rangle, \quad J^3 |\psi^c(\vec{p}, s_{1,2})\rangle = \pm \frac{1}{2} |\psi^c(\vec{p}, s_{1,2})\rangle$$

14 Exercise 14

Derive the result

$$\begin{aligned} (p+m) \left(\frac{1+\gamma^5 s_{\pm}}{2} \right) m \xrightarrow{\rightarrow} 0 p &\equiv P_{\mathbf{R}, \mathbf{L}} p \\ (p-m) \left(\frac{1+\gamma^5 s_{\pm}}{2} \right) m \xrightarrow{\rightarrow} 0 \frac{1 \mp \gamma^5}{2} p &= P_{\mathbf{L}, \mathbf{R}} p \end{aligned}$$

for the helicity projectors in the massless limit.

15 Exercise 15

Weak charged current transitions involve the chiral spinors $u_L(\vec{p}, s) \equiv P_L u(\vec{p}, s)$ and $v_L(\vec{p}, s) \equiv P_L v(\vec{p}, s)$. Show that in the relativistic limit such transitions mainly involve negative helicity particles or positive helicity antiparticle, and estimate the suppression factor for transitions involving the "wrong" helicity.

16 Exercise 16

Show in two ways that $|\bar{u}(\vec{p}_2, +)u(\vec{p}_1, -)|^2 = 2p_1 \cdot p_2$, where $u(\vec{p}, \pm)$ are the helicity spinors for massless fermion. The two ways are

1. directly from the form of the spinors in the chiral representation
 2. using trace techniques
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17 Exercise 17

Prove the Fierz identity

$$(\bar{w}_{1L}\gamma^\mu w_{2L})(\bar{w}_{3L}\gamma_\mu w_{4L}) = -(\bar{w}_{1L}\gamma^\mu w_{4L})(\bar{w}_{3L}\gamma_\mu w_{2L})$$

18 Exercise 18

Suppose a fermion ψ of mass m interact with a Hermitian scalar ϕ of mass μ with

$$\mathcal{L}_I = h\bar{\psi}\psi\phi$$

where h is small.

1. Calculate the spin-averaged differential cross section for $\psi(\vec{p}_1)\psi(\vec{p}_2) \rightarrow \psi(\vec{p}_3)\psi(\vec{p}_4)$ in the CM in terms of the invariants s, t and u .
 2. Specialize to $m = 0 = \mu$. Show that the scattering is isotropic in that limit and calculate the total cross section.
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19 Exercise 19

Consider $e^-(\vec{k}_1)\pi^+(\vec{p}_1) \rightarrow e^-(\vec{k}_2)\pi^+(\vec{p}_2)$ elastic scattering. Show that the spin averaged cross section in the pion rest frame is

$$\frac{d\bar{\sigma}}{d\cos(\theta_L)} = \frac{\pi\alpha^2 \cos^2\left(\frac{\theta_L}{2}\right)}{2k_1^2 \sin^4\left(\frac{\theta_L}{2}\right) \left[1 + \frac{2k_1}{m_\pi} \sin^2\left(\frac{\theta_L}{2}\right)\right]}$$

where θ_L is the scattering angle and $m_e \approx 0$.

20 Exercise 20

Calculate the CM differential cross section for the process $e^-(\vec{p}_1)\mu^+(\vec{k}_1) \rightarrow e^-(\vec{p}_2)\mu^+(\vec{k}_2)$ in terms of $s = E_{\text{CM}}^2$, the CM scattering angle θ , and the muon mass m_μ . We suppose the $m_e \approx 0$.

21 Exercise 22

Calculate the differential cross section for unpolarized Møller scattering $e^-e^- \rightarrow e^-e^-$, both in terms of the invariants and θ .
